

Jinsook Lee

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RESEARCH INTEREST

Socio-technical systems in education, computational social science, algorithmic decision-making, and societal impact of AI

EDUCATION

2022– present	Cornell University <i>Ph.D. in Information Science</i> Committee: René F. Kizilcec (co-chair), Thorsten Joachims (co-chair), Nikhil Garg
2025	<i>M.S. in Information Science</i>
	Korea University
2019	<i>M.S. in Fashion Marketing</i>
2017	<i>B.S. in Education</i>

PUBLICATIONS

JOURNAL ARTICLES & CONFERENCES

- P11 **Jinsook Lee** , Kirk Vanacore, Bakhtawar Ahtisham, Zhuqian Zhou & René F. Kizilcec (2026). Domain-Adapted Retrieval for In-Context Annotation of Pedagogical Dialogue Acts. *Under review for COLM2026* [\[arXiv\]](#)
- P10 **Jinsook Lee**, Conrad Borchers, AJ Alvero, Thorsten Joachims & René F. Kizilcec (2026). The Digital Divide in Generative AI: Evidence from Large Language Model Use in College Admissions Essays. *Learning at Scale 2026 (L@S2026)* [\[arXiv\]](#)
- P9 Kirk Vanacore, **Jinsook Lee**, Bakhtawar Ahtisham, Sarah Shaw, Justin Reich and Rene Kizilcec (2026). From Tutor Moves to Tutoring States: Modeling the Timing and Sequencing of Pedagogical Strategies for Student Engagement *Learning at Scale 2026 (L@S2026)* [\[arXiv\]](#)
- P8 Bakhtawar Ahtisham, Kirk Vanacore, Zhuqian Zhou, **Jinsook Lee** and Rene F. Kizilcec (2026). LLM Reasoning Predicts When Models Are Right: Evidence from Coding Classroom Discourse. *Educational Data Mining 2026 (EDM26)* [\[arXiv\]](#)
- P7 Zhuqian Zhou, Kirk Vanacore, Bakhtawar Ahtisham, Doug Pietrzak, Daryl Hedley, Jorge Dias, **Jinsook Lee**, René F. Kizilcec, Chris Shaw and Ruth Schäfer (2026). Utility-Preserving De-Identification for Math Tutoring: Investigating Numeric Ambiguity in the MathEd-PII Benchmark Dataset. *Educational Data Mining 2026 (EDM26)* [\[arXiv\]](#)
- P6 Bakhtawar Ahtisham, Kirk Vanacore, **Jinsook Lee**, Zhuqian Zhou, Doug Pietrzak & René F. Kizilcec (2026). AI Annotation Orchestration: Evaluating LLM verifiers to Improve the Quality of LLM Annotations in Learning Analytics. *Learning Analytics and Knowledge (LAK26)* [\[arXiv\]](#)
- P5 **Jinsook Lee***, Emma Harvey*, Joyce Zhou, Nikhil Garg, Thorsten Joachims & René F. Kizilcec (2024). Ending Affirmative Action Harms Diversity Without Improving Academic Merit. *AAAI/ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO' 24)*. [\[Paper\]](#) [\[Video\]](#) * equal contributions

- P4 **Jinsook Lee**, Yann Hicke, Renzhe Yu, Chris Brooks, & René F. Kizilcec (2024). The Life Cycle of Large Language Models in Education: A Framework for Understanding Sources of Bias. *British Journal of Educational Technology*, 55(5), 1982–2002. [\[Paper\]](#)
- P3 AJ Alvero, **Jinsook Lee**, Alejandra Regla-vargas, René F. Kizilcec, Thorsten Joachims, & antonio. (2024). Large Language Models, Social Demography, and Hegemony: Comparing Authorship in Human and Synthetic Text. *Journal of Big Data*, 11(1), 138. [\[Paper\]](#) [\[Press\]](#)
- P2 **Jinsook Lee**, Kibum Moon, Soyeon Han, Sukang Lee, Hyejung Kwon, & Jaeho Han (2021). Development and Application of an AI-Powered Adaptive Course Recommender System in Higher Education: An Example from K University. *Journal of Educational Technology*, 37(2), 267–307. [\[Paper\]](#) [\[Press\]](#)
- P1 Kibum Moon, Jinwon Kim, & **Jinsook Lee** (2021). The Early Prediction Model of Student Performance Based on Deep Neural Network Using Massive LMS Log Data. *Journal of the Korea Contents Association*, 21(10), 1–10. [\[Paper\]](#)

WORKSHOP & NON-ARCHIVAL CONFERENCE PUBLICATIONS

- W4 **Jinsook Lee**, AJ Alvero, Thorsten Joachims & René F. Kizilcec. (2025). Poor Alignment and Steerability of Large Language Models: Evidence from College Admission Essays. *Conference on Language Modeling (COLM25) SoLAR workshop, Social Sim' workshop* [\[arXiv\]](#) [\[Press\]](#)
- W3 **Jinsook Lee**, Yann Hicke, Renzhe Yu, Chris Brooks & René F. Kizilcec. (2024). When Bias Meets Personalization: Challenges and Perspectives in LLM-Based Educational Technology. *Learning Analytics and Knowledge (LAK24) Fostering Ethical and Equitable Learning Analytics workshop*
- W2 **Jinsook Lee**, AJ Alvero, Thorsten Joachims & René F. Kizilcec. (2024). Application Essays and Characters in Higher Education Admissions. *2024 Annual Meeting of the National Council on Measurement in Education (NCME24)*
- W1 **Jinsook Lee**, Bradon Thymes, Joyce Zhou, Thorsten Joachims, & René F. Kizilcec (2023). Augmenting Holistic Review in University Admission using Natural Language Processing for Essays and Recommendation Letters. *Artificial Intelligence in Education (AIED23) EDI in EdTech R&D Workshop* [\[arXiv\]](#)

BOOK CHAPTER

- B1 Sebastian Munoz-Najar Galvez, **Jinsook Lee**, AJ. Alvero, Sheridan Stewart, René F. Kizilcec, & Amy Desiderio (2025). Artificial Communication and Media Realism in College Admissions. *The Digitized Campus: Artificial Intelligence and Big Data in Higher Education* [\[Book\]](#)

WORK & RESEARCH EXPERIENCE

- 2025 **NYC FIRST Siegel PiTech Fellow**
Building Data Infrastructure for NYC FIRST: An Educational Third Space for STEM [\[blog\]](#)
Supervisor: Kate Karageorgiou
- 2022 **Seoul National University Research Fellow**
Built an AI-powered course/major recommender system using historical university enrollment and performance data

Developed an interactive recommendation dashboard and UI prototype using D3.js and Tableau
Supervisor: Young Hwan Cho, Ph.D.

2019–
2022 **Korea University** *Data Scientist*

Developed the first official AI-powered course/major recommender system in South Korea
MOU Project with K-MOOC to develop a course recommender system
Developed “Student interdisciplinary index” and created a dashboard across colleges
Analyzed online learning patterns and course survey results during COVID-19 using LMS dataset
Supervisor: Gyu-Tae Kim, Ph.D.

2019 **Arthur D. Little** *Intern*

Analyzed customer review text data using NLP methods to develop go-to-market strategies for a leading Korean motorcycle manufacturer

AWARDS & GRANTS

2025 **Siegel PiTech PhD Impact Fellowship**
2024 **Cornell Center for Social Sciences Grant**
2020 **Best Program Award** *Korea University*
2020 **Best Practice Nominee** *Tableau Korea*

INVITED TALKS

2023 Bias in Large Language Models in Education: Sources, Measures, and Mitigation Strategies
National Council on Measurement in Education-AI in Measurement and Education (NCME-AIMC)
2021 How We Upgrade ‘AI-advisor’ with Graph and the Future of Personalized Recommendation
Services in Higher Education *The 7th K.U. Innovation Forum, Seoul, South Korea* [\[video\]](#)
2020 Best Practice of University Innovation Case of Data Analysis with Tableau: Korea University
Tableau Korea Webinar [\[video\]](#)

TEACHING & SERVICE

TEACHING ASSISTANT

Cornell University

Language and Information (SP26)	Prof. Cristian Danescu-Niculescu-Mizil
Learning Analytics (SP25, SP24, SP22)	Prof. René F. Kizilcec
Designing Technologies for Social Impact (FA24)	Prof. Chris Csíkszentmihályi

ACADEMIC VOLUNTEER

Festival of Learning 2026 *Student Volunteer Chair*
AIED2026 *LBR Program Committee*

REVIEWER

ACM Conference on Fairness, Accountability, and Transparency (FAccT’25 & ’26)
Conference on Language Modeling Social Sim’ workshop (COLM25)
Journal of Learning Analytics (JLA)
International Journal of Artificial Intelligence in Education (IJAIED)

